

Village

Project Document

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INTRODUCTION

Aims and Objectives of the Project

“It takes a village. We are Village.”



The aim of our project is to develop an application that helps combat loneliness in parents by providing a platform to connect, share knowledge and exchange experiences about parenting. A [recent survey](#) revealed that 79% of respondents would value a way to connect with other parents, and 38% feel they have no one providing them with support in their role as a parent. The app's namesake, "Village" comes from the phrase *“it takes a village to*

raise a child” and we want to create and promote this community base for parents who need it.

Objectives:

- **Community:** Create a platform for parents to ask questions, discuss, and comment back.
- **Both online and in real life connection:** Provide information on local events based on parents’ search criteria. This encourages in-person connection (the app is focused on London for MVP).
- **Mental Health Support:** Offer resources for mental health as part of the forum and event offerings, as a key theme.
- **Filtering Capabilities:** Enable filtering by parent status (e.g. new parents, LGBTQIA+ parents, carers and single parents) to enable targeted interactions.

Roadmap of the Report

1. Background
2. Specifications and Design
3. Implementation and Execution
4. Testing and Evaluation
5. Conclusion

BACKGROUND

The project focuses on developing an app to support parents by providing:

- **A platform to connect and share experiences:** Parents can ask questions and share advice through forums. Community polls provide quick insights into relevant parenting topics, with immediate feedback via polling percentages.
- **A secure community and user profile:** the user has an account and logs in, so discussions are only visible by other members.
- **Local events and activities:** Customised based on parent’s search parameters and shown on a visually-appealing map.
- **Mental health resources and educational content:** Support groups and resources aimed at improving mental health, as well as a wealth of parenting topics.

Understanding the requirements:

- The target audience includes parents-to-be, new parents, and carers of babies and children.
- Parents need support to navigate the challenges of parenting, and Village can provide resources and education to support this.
- Village aims to combat loneliness by creating a safe and supportive community.

SPECIFICATIONS AND DESIGN

Technical Requirements

- **Webpages:** Six main pages including login, registration, user dashboard, events, forum and about us (‘meet the team’). A navbar and footer are included across all relevant pages.

- **Backend:** Node.js/Express, SQL database for data management (e.g. about users and forum posts) and API for interactions with the database. The app will also make use of the Ticketmaster API for searching, finding and displaying locations of events.
- **Frontend Tools:** React and Bootstrap for responsive design and Figma for UI/UX design.
- **Collaboration Tools:** Trello for project management, Miro for brainstorming, Slack and WhatsApp for communication and Git/Github for version control and remote repository storage.
- **Additional Features:** scope for development beyond MVP includes user notifications, language translation API interactivity, rating threads, searching the site/threads and behaviour/content moderation tools.

Non-Technical Requirements

- **User Experience:** Simple, intuitive interface with considerations for accessibility.
- **Diversity and Inclusion:** Use of diverse images, dyslexia-friendly fonts (Verdana, League Spartan), and contrast-tested colours to help those with colour blindness. Gender-neutral colours have been chosen to promote gender inclusion.

Colour Contrast Checker Results

To ensure accessibility, a colour contrast checker was used to evaluate the chosen colours. The results show that the combinations used in the app meet the required standards (tested on <https://colourcontrast.cc/>):



- **Button + Button Text:** *Login and Registration* - 7.06
- **Background Colour + Text:** *Login and Registration* - 14.04
- **Nav Bar Colour + Nav Item Menu Colour** - 6.87
- **Card Header + Header Text:** *Dashboard* - 6.21
- **Cards Background + Text:** *Dashboard and Events* - 7.76
- **Button Colour + Button Hover Colour:** *Dashboard and Events* - 19.11 - 21
- **Labels + Cards Background:** *Forum* - 8.91
- **Background + Text:** *Meet the Team* - 13.58

These choices ensure the application is accessible to users with visual impairments, making the content readable and the interface easy to navigate.

Design and Architecture



Draft Wireframes: Login, Sign Up and Dashboard

From the initial wireframes to the final implementation, we have taken into account the user experience. It was a priority for us to design a user-friendly app. We have designed an intuitive user flow experience that allows the user to focus on what really matters: connect with other parents. The overall user flow is: Sign Up > Login > Dashboard. From the Dashboard, the user can easily navigate to the other pages that are displayed in the navbar's Dashboard (Meet the Team, Forum and Events).

We agreed on an autumnal colour scheme (brown, mauve, green, orange, off-white) which is easy on the eyes and best fits the image of 'Village'.

1. Homepage: Login.

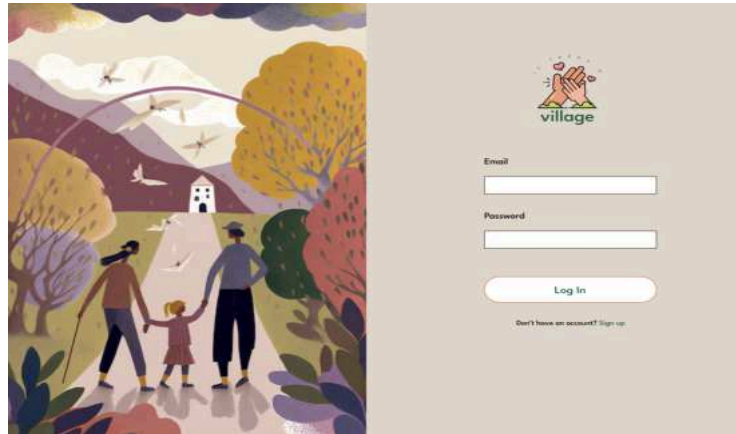
Login user flow: Login > Dashboard

The Login page will be the first page the visitor should see. To access the site and be a part of the exclusive Village community, they must have an account. This is done to protect the Forum and users from public view.

The user needs to provide the registered email and password to log in. If the email and password don't match with a record stored in the database, an error is shown indicating that the password or email is invalid.

Redux is used to globally manage the state related to user authentication, including login and logout. The **jsonwebtoken** library is used for user authentication making the login process more secure.

After clicking on the Sign In button, the user is redirected to the Dashboard. In the navbar, there is a 'Log Out' button that only renders if the user is logged in. This component uses the 'useSelector' hook to access the user state from the Redux store.



Draft Wireframe: Login Page



Final Login Page

2. Sign Up – User Registration

Sign Up user flow: Sign Up > Login > Dashboard

User clicks ‘Sign Up’ below the Sign In form to create an account, leading the user to the registration process. Here, they input account registration details: *first name, last name, username, birthday, email, postcode, location, password and repeated password*.

The form is placed on the right side of the page to maintain a consistent format and to not disrupt the user’s attention away from it (the previous page has a form on the right side). Instead of following the initial wireframe design with a registration roadmap on the left, it was decided to feature an image as it makes the page more welcoming.

During the registration process the password is hashed before being saved in the database, using **bcrypt**. All fields of the registration form must be completed. There are some validations used:

- the name and surname must each have more than 2 letters,
- the password and repeated password fields must match,
- and a regex is used so the user provides a sufficiently secure password.

When errors arise, these are shown to the user below the appropriate fields for correcting them. In addition, if the email or username already exists in the database, the user cannot register and an error message is shown.

Draft Wireframe: Sign Up Page

Final Sign Up Page

After the user inputs their important details and clicks on the Sign Up button, the user is redirected to the login page.

The 'Login In' link provided below the form will pivot the user back to the login page if they already have an account.

Once signed up and signed in, they are now a 'villager'.

3. User Dashboard

Dashboard user flow: Login > Dashboard > links to Recent Forum Posts/Image Carousel/Broadcast/Poll/Events

After logging in, users see the dashboard, which serves as a gateway to various app elements, unifying them for a cohesive experience.

Event Features

Event features are displayed on the right side of the dashboard. Events Spotlight highlights four popular events in London for villagers to attend that week. Events on Sale Now allows users to explore future events with newly available tickets. These components differ from the original wireframe, as "Past Events," "Upcoming Events," and "Events Near You" were removed to enhance user experience and reduce unnecessary information.

Village Poll

The village poll feature fosters community among users. By voting on topics, users gain insight into community opinions, providing us with data to optimise the platform. Votes are currently anonymous, but the results table's structure allows easy implementation of user identification if needed in the future. The poll's design was simplified from the original wireframe to gauge user interest before enabling them to post their own questions.

Recent Forum Posts

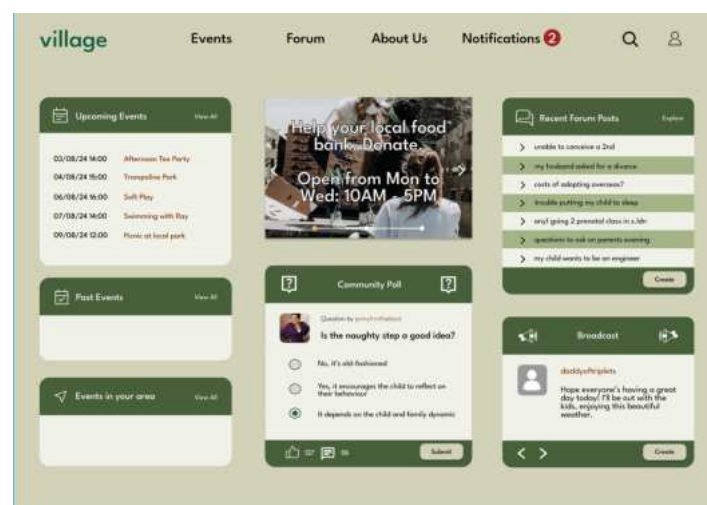
Recent Forum Posts fosters personal engagement by directing users to the forum. It displays the seven most recently updated threads, preventing frequent comments from dominating and ensuring diverse content. Each topic is clickable, routing users to the thread and displaying the date of the most recent post. The option to create and explore the forum, shown in the wireframe, was removed to avoid redundancy since the forum can be accessed via the navbar.

Broadcast Feature

Users can create messages by clicking 'Create'. Fellow villagers can see who posted the message and use arrows in the footer to view other messages. Broadcast messages expire after 24 hours, preventing the component from becoming overwhelmed.

Image Carousel

The image carousel is at the top centre of the dashboard, showcasing three images to inspire users to explore different experiences in London. The image changes when users click one of three small dots below it. It's a React responsive carousel with easily updatable images. Unlike the original wireframe, the text is subtly placed beneath the images rather than overlapping them.



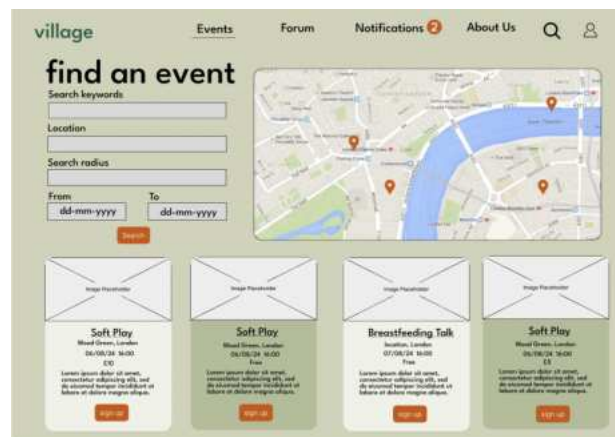
Draft Wireframe: User Dashboard



Final User Dashboard Page

4. Events Page:

The Events page will list all family-friendly events in London. The villager can search on various criteria (keywords, location, radius, datetime range) to find events they wish to attend. Clicking 'search' will surface these events on the lower half of the screen. The form will not validate if they input a 'from' date value later than the 'to' date value. They can then click on the respective card to go to the external event booking web page to checkout. The event card shows the event's name, address, date and time, ticket price, and a link that leads the user to book tickets on Ticketmaster.



Draft Wireframe/Design: Events Page

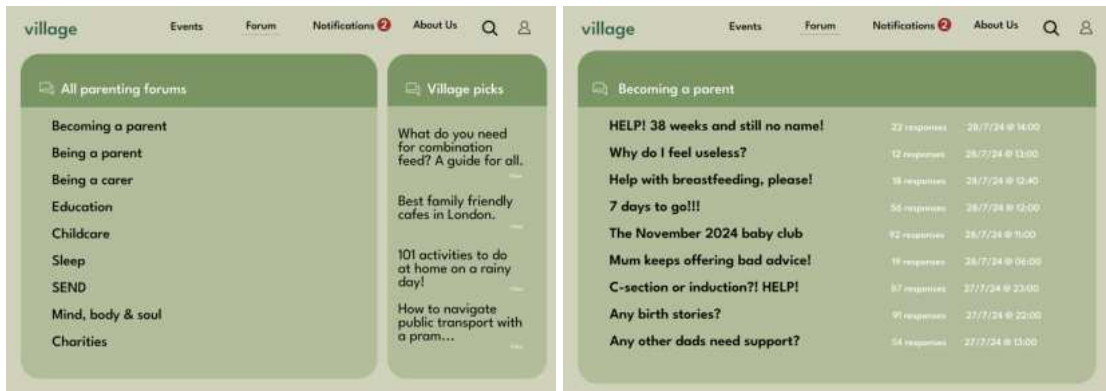


Final Events Page

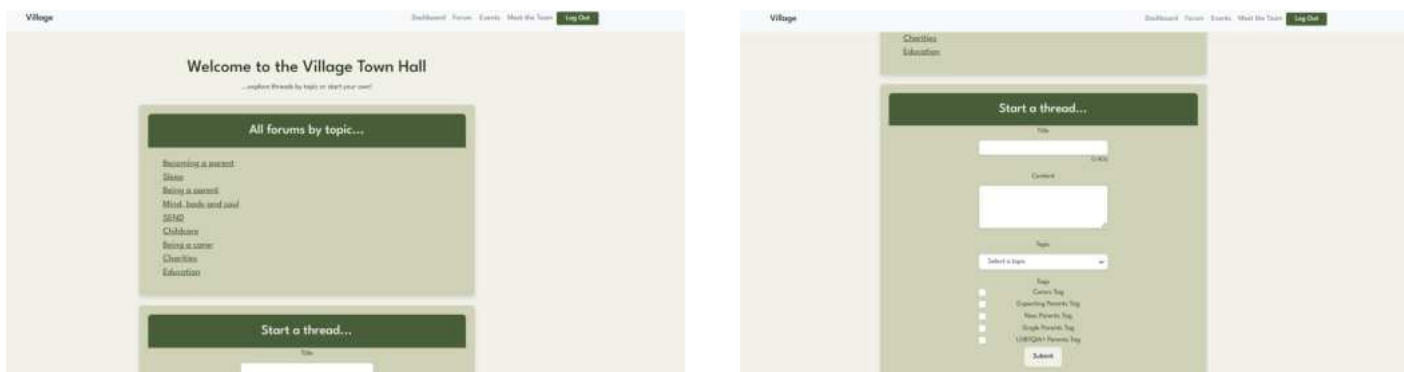
5. Forum Pages:

Forum user Flow: All parenting forums by topic & form to create new thread > Link into threads by topic which can be filtered by tags > into the singular thread with reply functionality

Platform for discussion and support: the Village parenting forums show forums at the top level, organised by topic and can be filtered by tags. Users can start a conversation, reply to a thread and explore other forums.



Forum Pages from wireframe



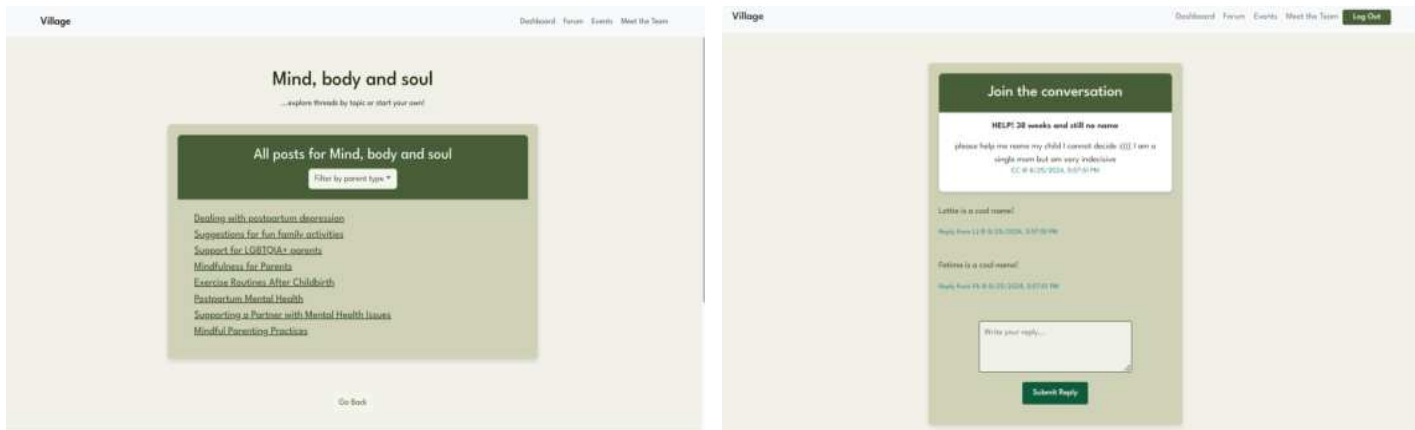
Main forum screen

All forums by topic

Here the user can access the entire database of posts listed by topics. Back button functionality allows the user to navigate back up to the previous level.

Start a thread

This posts a new thread to the forum and will then appear in the topic chosen.



Forum pages from app

Topics

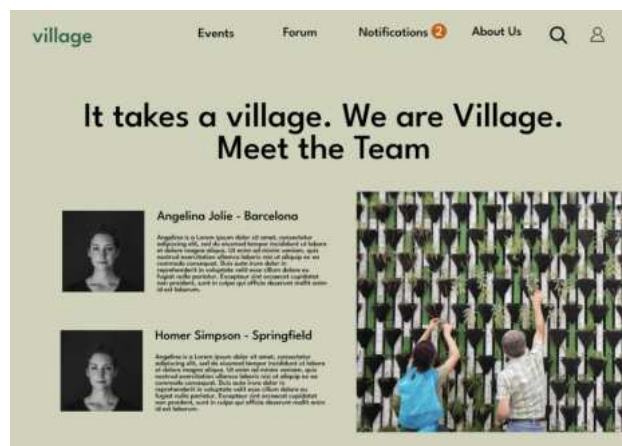
This screen contains each thread filtered by topic, which the user can then filter further by parent type, gaining access to threads that suit them and their family.

Individual thread

This shows threads individually where the user can browse and reply, enabling them to join in the conversation and participate in the community.

6. Meet the Team:

An information page to introduce the team members behind Village.



Daft wireframe: Meet the Team



Final page: Meet the Team

The user is able to scroll through the team members' headshots and bios. Bootstrap means the design on this page is fully responsive.

- **UI/UX Considerations:** Consistent navigation, easy-to-read fonts, and a colour palette that is pleasing and accessible.

Architecture

- **Frontend:** Bootstrap for a responsive and consistent design. Figma for creating detailed wireframes and UI elements.
- **Backend:** Node.js for server-side logic. SQL database for managing user data, event listings, and forum posts.
- **API Integration:** Integration with Ticketmaster API for event information, and SQL database interactions for backend data management to facilitate user information and forum interactions.

IMPLEMENTATION AND EXECUTION

Development Approach and Team Member Roles

- **Agile Development:** Adopted for iterative progress and continuous improvement with week-long sprints, retrospectives, regular standups and longer planning meetings.
- **Team Roles:**
 - Roles looking after the Trello board, meeting coordination, Miro, documentation, Figma, testing, and the final presentation have been shared amongst the team.
 - Screens have also been assigned amongst mini 'teams' for coding up the first batch of screens in initial sprints.

Tools and Libraries

- Frontend Tools: React, Bootstrap, Figma, Jest, dayjs, Axios, react-redux, @reduxjs/toolkit.
- Backend Tools: Node.js, Express, MySQL, jsonwebtoken, bcrypt, cors.
- Collaboration Tools: Trello, Slack, GitHub, GoogleDocs, Miro, WhatsApp.

Implementation Process

Approach

After an all-hands approach to brainstorming (using Miro, pictured), we broke down features into 'must-have' and 'nice-to-have' to allow prioritisation for MVP, before detailing the various frontend and backend requirements screen by screen.

When planning, we split out the screens of our functionality into backend and frontend to understand the finer details of the requirements needed. For example, use of get and post requests to API for interaction between the forum and database.

We then split into smaller groups to tackle screens:

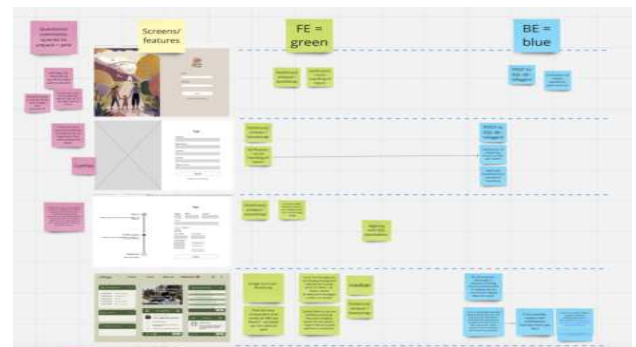
- Cynthia/Maria: login and registration
- Anh/Chloe: dashboard and events
- Cat/Lottie/Fatima: forum, meet the team, navbar, footer.

Through a mixture of more brief standups and longer planning meetings, the team updated on progress, reshuffled priorities, made decisions and slowly but steadily built a village!



Trello was used to indicate to do / doing / done / blocked. This was referenced in standups to understand capacity and progress.

We have each taken it in turns to review pull requests and feedback to one another to ensure continual functionality of the app.

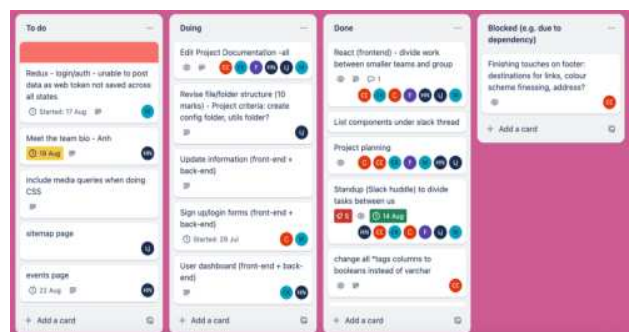


Achievements

Initial Achievements

- Initial Wireframes and Prototypes: Developed and reviewed for user feedback.
- Backend Setup: An Express API using mysql2.
- Forum and Event Listings: researched key functionalities for user interaction and event information, using ticketmaster public API and our own Database API.

Final Achievements:



- Developing 6 screens for the Village project: Sign In, Log In, Dashboard, Forum, Events and Meet the team, as well as a navbar and footer
- Implementing responsive design for the screens
- Using hashing for passwords in the database during the registration and log in processes
- Establishing authentication for login with JSON Web Tokens
- Using regex validation during registration to make passwords more secure
- Using Redux to globally manage the state related to user authentication, including login and logout

- Effective remote collaborate using new technologies to deliver an ambitious MVP scope
- Comprehensive unit testing environment using Jest, React Testing and Isometric Fetch as polyfilling.

Challenges

- **Real-time Updates:** Implementing efficient mechanisms for live interactions in forums.
- **Getting accustomed to Git/GitHub group workflow:** for most of us, this was the first time we worked as a group using GitHub and opening and reviewing pull requests. We managed to familiarise ourselves with the dynamics of group work and Git/GitHub during our first week of code as the group held standups every 2-3 days and communicated daily on WhatsApp.
- **Ticketmaster API's definition of family-friendly:** extremely hard to filter unique family-friendly events despite the includeFamily parameter that does enable this; not all events on Ticketmaster were defined family-friendly or not, hence the lack of varied results. In place of this, the keyword, children, was used to filter for family-friendly events.
- **Ticketmaster API - duplicate events:** additionally, the same exact event happening on different days were given different event IDs, so to showcase a unique list of events instead of duplicate events on different days, I filtered events by name in client/front-end. That is why we were only able to show events from a smaller time window and far less results than what is stated in the backend (size: 200, routes/events.js). However, using the form will help filter for events past the general time frame displayed on the event cards (events page).
- **Unable to grab user's token/information from centralised state:** due to limited time, we were not able to link user token or credentials to our code that would enable the currently logged in user to post instead of posting as the default user. We understand Redux may help with this and given additional time we would have accomplished this.

Agile Development

- **Iterative Approach:** Regular sprints for incremental progress.
- **Code Reviews and Testing:** Frequent evaluations to ensure quality and functionality.

TESTING AND EVALUATION

Testing Strategy

Functional Testing

- **Objective:** Unit testing on components to verify they work as intended using Jest, React Testing library and Postman.
- **Scope:** Includes user registration, event listings, forum discussions, and filtering capabilities

User Testing

- **Objective:** Collect feedback from target users to improve usability and functionality.
- **Methods:** Beta testing with a selected group of parents to gather insights and suggestions.

System Limitations

- **Scalability:** Potential challenges as the user base grows, requiring efficient backend management.
- **Performance:** Continuous updates and maintenance necessary to handle increasing data and user interactions.

Examples of Village Testing:

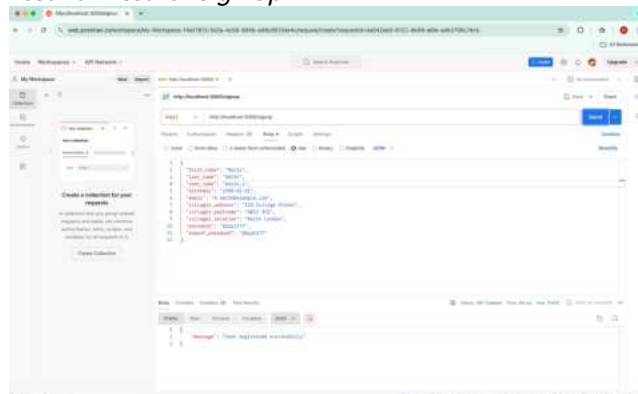
Unit testing with Jest:

```
> vite-project@0.0.0 test
> jest

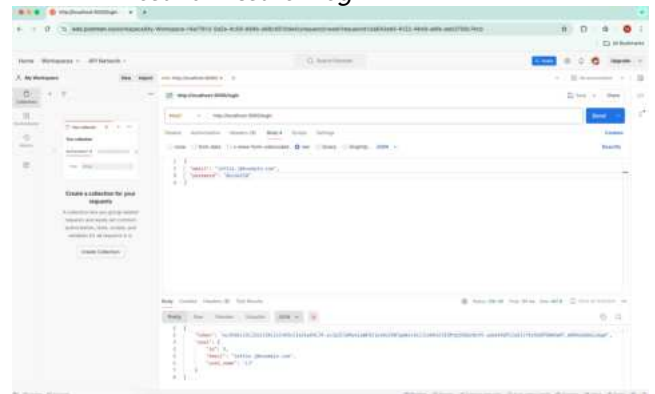
PASS src/components/logoutbutton/LogoutButton.test.js
PASS src/components/dashboard/ImageCarousel.test.js
PASS src/components/registration/Registration.test.js
PASS src/components/login/Login.test.js
PASS src/components/dashboard/Card.test.jsx
PASS src/components/dashboard/Poll.test.js
PASS src/components/forum/forum.test.js
PASS src/components/navbar/Navbar.test.js
PASS src/app.test.js
PASS src/components/dashboard/Broadcast.test.js

Test Suites: 10 passed, 10 total
Tests: 21 passed, 21 total
Snapshots: 0 total
Time: 2.042 s
Ran all test suites.
lottiejudge@lotties-air client %
```

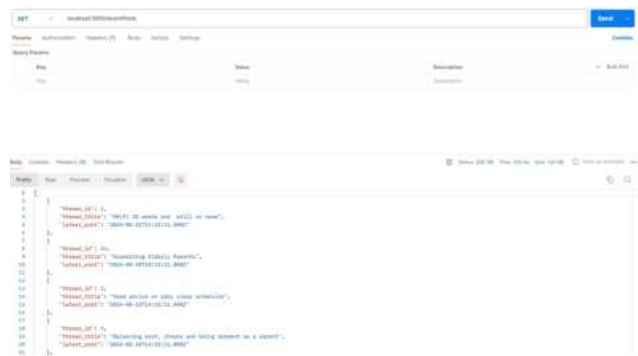
Postman Test for Sign Up



Postman Test for Log In



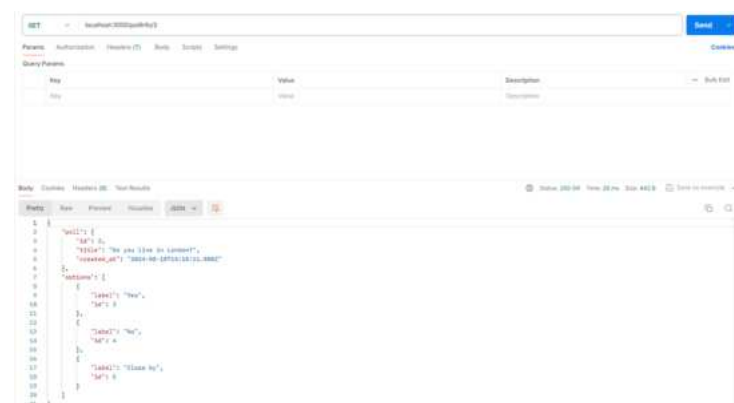
Postman Test for Recent Forum Posts Component



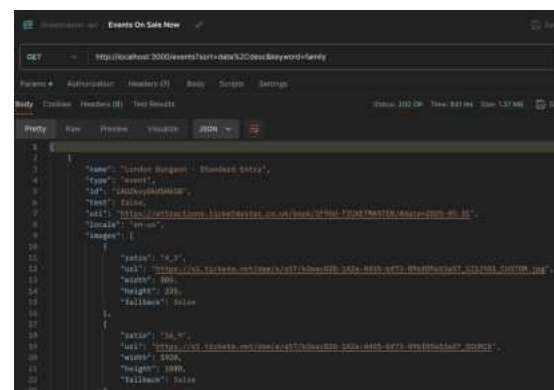
Postman Test for Poll Results



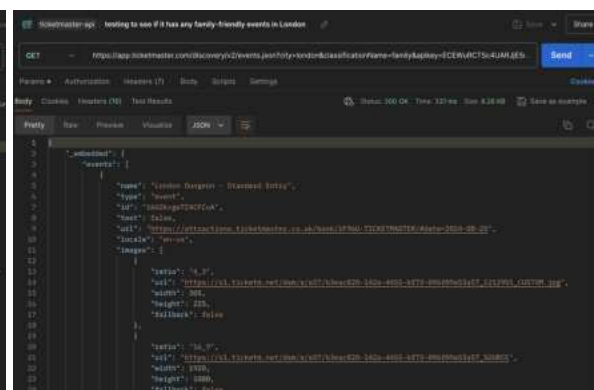
Postman Test for Poll Information



Postman Test for Events



Ticketmaster API - check if we're able to grab events



CONCLUSION

We have followed an Agile methodology to develop the React app 'Village'. We divided the work into smaller tasks, assigning specific screens to sub-groups, and met regularly to make collective decisions, identified priorities, and reorganised work as necessary. This flexible approach has enabled us to successfully develop six pages for the Village app. Thanks to

the group project we have learned a lot about team collaboration and full-stack development.

The app is a responsive platform designed to help parents connect. We have prioritised user experience by creating an intuitive user flow and ensuring the app is also visually appealing. For styling, we have used Bootstrap and CSS to achieve an elegant and cohesive design. Security was also a top priority: we implemented password hashing and user authentication to protect user data.

The Dashboard serves as a central hub, offering an overview of all Village features at a glance, including events, polls, forums, broadcast messages and other resources. From the Dashboard page, parents can easily navigate to the Forum and Events pages, which are key to fostering connections among users. The Forum provides a space for parents to share experiences and knowledge on various aspects of parenthood. Meanwhile, the Events page allows parents to discover activities where they can connect with other families. For the Events page, we have integrated the Ticketmaster API to fetch relevant events based on user input.

Testing has been an important part of our project. We have used Jest and Postman to test the app and its components, ensuring it works as intended.

The project aimed at creating a supportive community for parents by addressing loneliness through connectivity and resource sharing: fostering online and offline mutual support.

The app's design and architecture has focused on user-friendly and functional features and inclusive content to cater to a diverse audience with a targeted need. Regular testing and agile practices ensured the app remains functional, scalable, and user-centric.

Welcome to Village 🍷.

APPENDICES

Appendix 1 - Image bibliography

Image	Location	Credits	Url
	'Meet the Team'	Person planting on hanged pots by Daniel Funes Fuentes	https://unsplash.com/photos/person-planting-on-hanged-pots-TyLw3IQALMs
	'Registration'	Woman holding a kid at the street, by Sai de Silva	https://unsplash.com/photos/woman-holding-kid-at-the-street-httxBNGKapo
	'Dashboard'	Field of pumpkins by Marius Ciocirlan	Squash scatter on the ground photo – Free Pumpkin Image on Unsplash
	'Dashboard'	Cookies by Rai Vidanes	Cookies on black grill beside yellow ceramic cup photo – Free Baking Image on Unsplash
	'Dashboard'	Yellow London by Ming Jun Tan	Yellow flowers photo – Free London Image on Unsplash

